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based on shared apomorphic characters is called **cladistics** (from Greek, *klados*-branch).

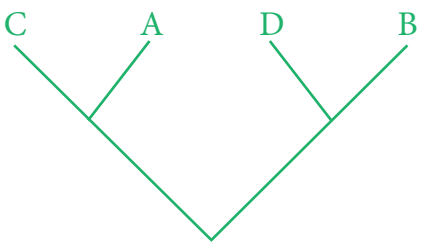
The outcome of a cladistic analysis is a **cladogram**, a tree-shaped diagram that represent the best hypothesis of phylogenetic relationships. Earlier generated cladograms were largely on the basis of morphological characters, but now genetic sequencing data and computational softwares are commonly used in phylogenetic analysis.

Cladistic analysis

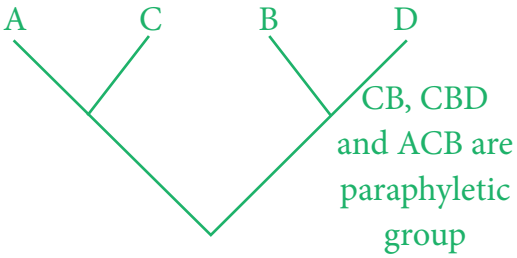
Cladistics is one of the primary methods of constructing phylogenies, or evolutionary histories. Cladistics uses shared, derived characters to group organisms into clades. These clades have atleast one shared, derived character found in their most recent common ancestor that is not found in other groups hence they are considered more closely related to each other. These shared characters can be morphological such as, leaf, flower, fruit, seed and so on; behavioural, like opening of flowers nocturnal/diurnal; molecular like, DNA or protein sequence and more.

Cladistics accept only **monophyletic groups**. **Paraphyletic** and **polyphyletic** taxa are occasionally considered when such taxa conveniently treated as one group for practical purposes. Example: dicots, sterculiaceae. Polyphyletic groups are rejected by cladistics.

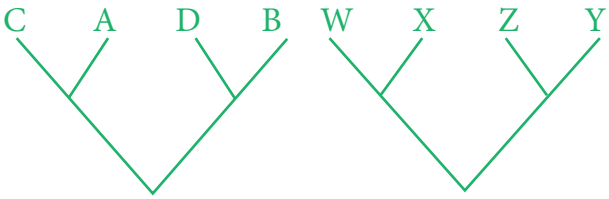
i. **Monophyletic group;** Taxa comprising all the descendants of a common ancestor.



ii. **Paraphyletic group;** Taxon that includes an ancestor but not all of the descendants of that ancestor.



iii. **Polyphyletic group;** Taxa that includes members from two different lineages.



Need for cladistics

1. Cladistics is now the most commonly used and accepted method for creating phylogenetic system of classifications.
2. Cladistics produces a hypothesis about the relationship of organisms to predict the morphological characteristics of organism.
3. Cladistics helps to elucidate mechanism of evolution.

5.13 Selected Families of Angiosperms

Dicot Families

Plant kingdom is so vast and varied. For the purpose of study, they have been classified into Artificial, Natural, Phylogenetic and APG system in course of time. Bentham and Hooker system of classification is followed in India till recently. Great variation occurs not only in different families, but also varies in genera and species